

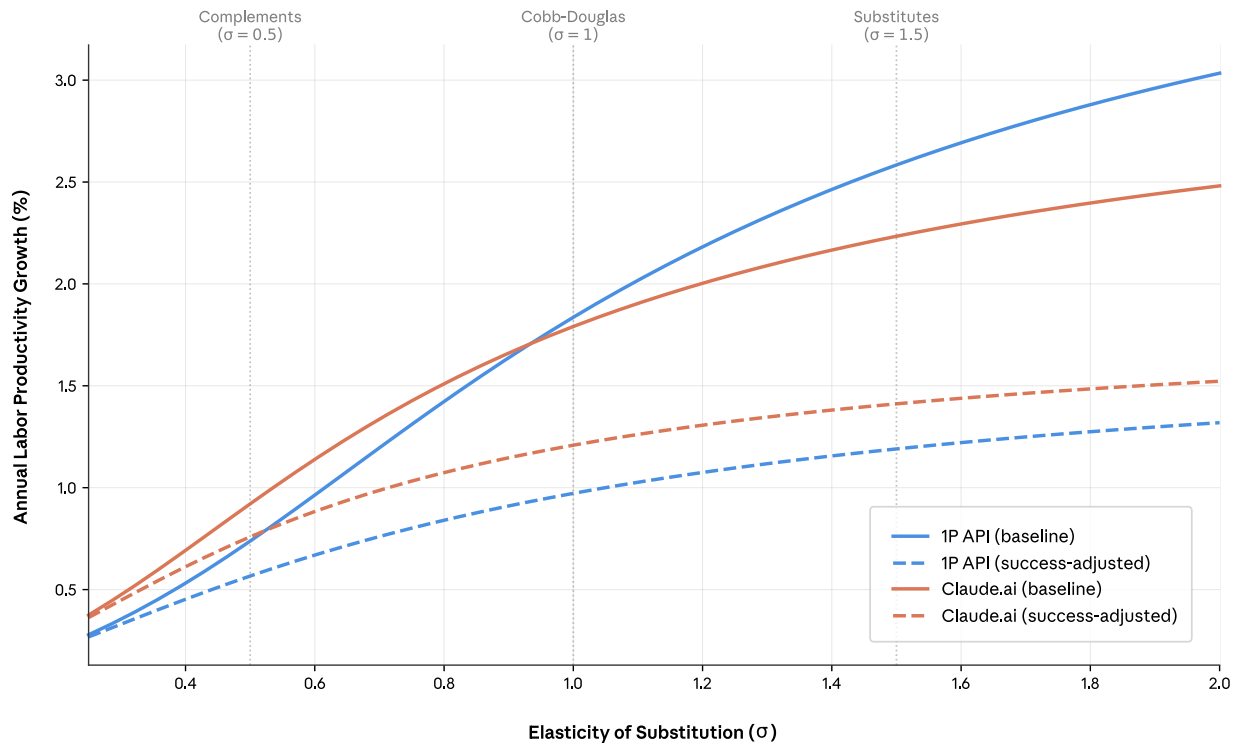
lesson plans more efficiently with AI while having no impact on time spent with students in the classroom.

To operationalize this idea, we impose some structure on how we aggregate task-level time savings within occupations but otherwise add up occupational efficiency gains as in the main analysis. Specifically, we suppose that within each occupation tasks are combined according to a Constant Elasticity of Substitution (CES) aggregator, where each task is weighted by the estimated time spent on each task as calculated in our [earlier analysis of the productivity effects implied by Claude usage](#).<sup>9</sup>

The key parameter is the elasticity of substitution across tasks,  $\sigma$ . When the elasticity of substitution is less than one, tasks are complements and those tasks that are not sped up by AI become bottlenecks for broader productivity gains. Alternatively, when the elasticity of substitution is greater than one, then workers can allocate toward the more productive tasks—thereby amplifying the overall time savings at the occupational level. An elasticity of substitution equal to one is a special case that replicates the main analysis above.

Figure 4.6 reports the results of this exercise for different values of task substitutability. As expected, when the elasticity of substitution is equal to one the implied productivity effect is the same as in our baseline analysis: An increase in labor productivity growth of ~1.8 percentage points per year over the next decade implied by both Claude.ai and API samples.

## Productivity impact by task substitutability



**Figure 4.6 Implied labor productivity effect from AI as a function of within-occupation task substitutability**

This figure shows the implied aggregate labor productivity growth over the next decade based on efficiency gains estimated for tasks with at least 200 observations in our sample of 1M conversations on Claude.ai and 1M records from 1P API traffic. The elasticity of substitution governs how the degree to which non-AI enhanced tasks constrain the occupational productivity gains implied by Claude usage under a model in which occupational output is a CES index across tasks. An elasticity of  $\sigma = 1$  reproduces our unadjusted baseline result of 1.8 percentage point increase in labor productivity growth over the next decade. Success-adjusted curves discount task-level speedups by task reliability. See text for more details.

When tasks are complements, however, the implied aggregate labor productivity impact declines sharply as the economic effects are bottlenecked by tasks that AI speeds up the least. For example, at  $\sigma = 0.5$  the implied overall labor productivity effect is 0.7–0.9 percentage points per year—around half the size as implied by our baseline estimates. Additionally adjusting for task success further reduces the implied productivity effects to 0.8pp for Claude.ai and 0.6pp for API.

On the other hand, when the elasticity of substitution is greater than one, the implied labor productivity based on pre-Opus 4.5 usage patterns is materially higher. For example, at  $\sigma = 1.5$  the implied labor productivity effect rises to

2.2–2.6 percentage points per year, consistent with greater specialization in tasks where AI provides the largest speedups.

In both cases the implied productivity impact based on API traffic is more responsive to the degree of task substitutability. This is consistent with the fact that there is a larger share of API traffic concentrated in fewer tasks and associated occupations as compared to Claude.ai: When tasks are complements, this concentration amplifies the bottleneck problem; when they are substitutes, it amplifies productivity gains from task specialization.

What this analysis shows is that the productivity effects of automation may ultimately be constrained by bottleneck tasks that elude AI automation for the time being. And the labor market implications of increasingly capable AI could be similarly affected by such forces. For example, [Gans and Goldfarb \(2026\)](#) argue that the presence of bottleneck tasks within jobs means that partial AI automation can lead to an increase in labor income as such tasks increase in economic value (at least until a job is *entirely* automated).

## Conclusion

The upshot of this chapter is that the impact of AI on the economy is unlikely to be uniform. As our effective AI coverage framework illustrates, the labor market implications for different workers will hinge on how reliable frontier AI tools are for their most central tasks.

But the labor market effects may also depend on the skill requirements of tasks that AI can proficiently handle relative to the rest of the economy. Indeed, we find that removing tasks Claude can already handle from the economy would produce a net deskilling effect: the tasks remaining for humans have lower educational requirements than those handled by AI.

While highly suggestive, this may miss an important detail: the most complex tasks where Claude is used tend also to be those where it struggles most. Rather than displacing highly skilled professionals, this could instead reinforce the value of their complementary expertise in understanding AI's work and assessing its quality.

The counterpart to these transformative labor market effects is the broader impact on growth and productivity. On the one hand, incorporating task reliability into our analysis diminishes the implied effect on labor productivity growth as informed by current Claude usage patterns. If bottleneck tasks bind, the implied impact diminishes further. On the other hand, the continuing growth in model capabilities suggests that both task coverage and task success may increase, which, in turn, could increase productivity impacts.

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<sup>1</sup> When we study the correlation between primitives with the O\*NET, we restrict to tasks appearing in at least 100 conversations to reduce measurement error. In the coverage analysis, we use all tasks above the privacy threshold of 15.

<sup>2</sup> Our online appendix is available at <https://huggingface.co/datasets/Anthropic/EconomicIndex>.

<sup>3</sup> See also [Tomlinson et al \(2025\)](#) for a related AI applicability score.

<sup>4</sup> We generate embeddings for each task statement using a pretrained sentence transformer (all-mpnet-base-v2) and predict education with Ridge regression.

<sup>5</sup> On the other hand, some historical evidence suggests that when technologies automating job tasks appear in patent data, employment and wages subsequently fall for exposed occupations ([Webb 2020](#)).

<sup>6</sup> When we first assessed the aggregate productivity implications of Claude usage, we relied on a sample of 100k Claude.ai conversations from Fall 2025. Based on the set of tasks for which we observed speedups, we estimated that labor productivity could be 1.8 percentage points higher per year over the next decade. Expanding the sample to 1M observations means that we need to take a stand on how to handle very infrequently occurring tasks—which are very common given that usage follows a power law, as we documented in our past report. We choose a threshold of 0.02% because it replicates our previous results for our sample of Claude.ai conversations. For privacy-preserving reasons, we only ever analyze tasks with at least 15 observations, or an implied threshold of 0.015% for a 100k sample. And so our results are internally consistent across samples. If we do not impose a restriction on our